

Lesson 28 — Strike a Pose (train a custom classifier)

Matches LEGO Teacher Portal lesson: "Strike a Pose".

Objective (I can...): I can create and train my own classifier by giving it example photos. **CS concept:** Training a custom classifier (examples become training data) · **Builds on:** Lesson 27 (using classifiers). *Privacy: photos are captured on-device only, used just for training, and never saved or uploaded.*

AI Connection (2 min): Now YOU are the teacher — you show the AI examples so it learns, the payoff of every "AI learns by repeating examples" note this year. *First-principles question:* "How many examples of each pose do you think it needs to tell them apart?"

Materials: LEGO CS & AI kit + device per pair with camera, open floor space to pose. **New build for the train-your-own sub-unit** — build the Strike a Pose reactor model now and keep it assembled through Lesson 29.

60-minute flow - Warm-up (8): "Teach a friend" — one student describes a pose in words; partner copies. More clear examples make learning easier — same for AI. - **Concept + demo (10):** Explain training: show the classifier several photos of each pose so it learns to tell them apart. Demo capturing a few examples for two poses. Restate: images stay on-device, not saved. - **Build/Code (30):** Pairs **build the pose-reactor model**, create a custom classifier with 2 pose categories, photograph several examples of each themselves, then program the model to react to each recognized pose. **Examples are data:** on their plan sheet, pairs keep a simple tally of examples-per-category (reusing the counting/Variables skill) and ask "are my categories balanced?" *Swap at 15 min so both capture examples and code.* - **Challenge (7):** Add a third pose category; test whether the model still tells them apart. - **Share + pack-up (5):** Pairs demo their trained classifier recognizing their poses. **Leave the model built.**

Pair roles: Driver captures/labels example photos, Navigator codes the reactions. Swap at 15 min. (*Trio rotates pose/capture/code.*)

Differentiation: *Support* — start with two very different poses. *Extension / fast-finisher* — use subtle poses and add more examples. *Reteach* — capture examples together and watch the label appear correctly before coding.

Exit check: Pair trains a custom classifier that recognizes their own poses and triggers actions. **At-home (optional):** Practice 3 clear poses; think about how you'd describe each. ~10 min.